

The Program in Action

When you run *towerofhanoi.py*, the output will look like this:

The Tower of Hanoi, by Al Sweigart al@inventwithpython.com

Move the tower of disks, one disk at a time, to another tower. Larger disks cannot rest on top of a smaller disk.

More info at https://en.wikipedia.org/wiki/Tower_of_Hanoi

```
      ||           ||           ||
      @_1@         ||           ||
      @@_2@@        ||           ||
      @@@_3@@@       ||           ||
      @@@@_4@@@@      ||           ||
      @@@@@_5@@@@@    ||           ||
                   A      B      C
```

Enter the letters of "from" and "to" towers, or QUIT.
(e.g. AB to moves a disk from tower A to tower B.)

> ab

```
      ||           ||           ||
      ||           ||           ||
      @@_2@@        ||           ||
      @@@_3@@@       ||           ||
      @@@@_4@@@@      ||           ||
      @@@@@_5@@@@@    ||           ||
                   A      B      C
      @_1@
```

Enter the letters of "from" and "to" towers, or QUIT.
(e.g. AB to moves a disk from tower A to tower B.)

--snip--

How It Works

The data structure that represents a tower is a list of integers. Each integer is the size of the disk. The first integer in the list represents the bottom disk, and the last integer represents the top disk. For example, [5, 4, 2] would represent the following tower:

```
      ||
      ||
      @@_2@@
      @@@@_4@@@@
      @@@@@_5@@@@@
```

Python's `append()` and `pop()` list methods can add and remove values from the end of the list, respectively. Just as `someList[0]` and `someList[1]` allow us to access the first and second values in a list, Python lets us use negative indexes to access values from the end of the list using expressions